VANSH BARASWAL

Full-Stack Developer · AI/ML · DSA

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SUMMARY

B.Tech IT student at HBTU Kanpur with strong fundamentals in data structures, algorithms (350+ LeetCode problems solved), and full-stack development. Hands-on experience in concurrent systems, multithreading, and distributed architecture. Hackathon winner at PW IOI Noida 2026. Experienced in building end-to-end web applications using the MERN stack and Django, with working knowledge of Azure cloud services and TypeScript. Active open-source contributor passionate about scalable systems and AI-integrated software.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, C++, SQL
Frontend: React.js, HTML5, CSS3, Tailwind CSS
Backend: Node.js, Express.js, Django, REST APIs
Cloud & Tools: Microsoft Azure (Fundamentals), Git, GitHub, Postman, VS Code, Netlify, Render
Databases: MongoDB, SQL
CS Fundamentals: Data Structures & Algorithms (350+ LeetCode — medium/hard focus), OOP, DBMS, Concurrency & Multithreading, Distributed Systems (basics)

EXPERIENCE

Software Engineering Virtual Intern — *JPMorgan Chase & Co. (Forage)* 2025
- Completed JPMorgan's SWE simulation — set up a dev environment, fixed broken data feeds, and built a live data dashboard using React.
- Worked with real-world financial data pipelines, applying REST API integration and frontend component architecture.

Software Engineering Virtual Intern — *Tata Consultancy Services (Forage)* 2025
- Completed TCS's engineering simulation covering software design, testing, and delivery workflows in an enterprise environment.
- Practiced requirement analysis, solution design, and professional technical documentation.

Open Source Contributor — *GirlScript Summer of Code (GSSoC)* 2026
- Actively contributing to open-source repositories — submitting PRs, resolving issues, and collaborating with maintainers across the community.

ML Research Contributor — *GridSage AI — Team Research Project* 2025
- Collaborated in a 6-member team to build a time-series electricity demand forecasting system achieving ~85% prediction accuracy on synthetic datasets.
- Owned data preprocessing and feature engineering pipeline; applied train-test splitting and ML evaluation metrics.

PROJECTS

GridSage AI — Electricity Demand Forecasting | *Python · Pandas · NumPy · ML* 2025
- Built a time-series forecasting model on hourly electricity demand data with seasonal and peak-hour pattern recognition.
- Engineered features from raw synthetic datasets, achieving ~85% prediction accuracy using regression-based ML models.
- Contributed to a 6-member research team; authored key sections of the technical research documentation.

Student Management System | *Django · HTML5 · CSS3 · Git* 2025
- Developed a full-featured web application supporting management of 500+ student records with CRUD operations.

- Implemented server-side form validation, Django ORM queries, and role-based data access.
- Maintained clean version history using Git with feature-branch workflow.

Task Manager — Full-Stack To-Do App | *React.js · Node.js · Express.js · MongoDB* 2025

- Designed and deployed a full-stack task management app with React frontend, Express REST API backend, and MongoDB persistence.
- Implemented JWT-ready routing, real-time UI state updates with React hooks, and responsive Tailwind CSS design.
- Deployed backend on Render and frontend on Netlify with CI-style manual deployment pipeline.

Concurrent Task Scheduler & Producer-Consumer System | *Python · Threading · Synchronization · System Design* 2026

- Designed and implemented a multi-threaded task scheduler in Python using thread pools, locks, and semaphores to safely coordinate concurrent workers across shared queues.
- Built a classic producer-consumer pipeline with bounded buffer, condition variables, and deadlock-free synchronization handling up to 10 concurrent producer and consumer threads.
- Documented system architecture covering concurrency model, thread lifecycle, and failure-recovery strategy; reduced task execution latency by ~40% vs. sequential baseline.

MERN Practice Suite | *React.js · Node.js · Express.js · MongoDB* 2025

- Built multiple end-to-end mini-apps to master component architecture, REST API consumption, and full deployment cycle.
- Explored authentication flows, error handling middleware, and MongoDB schema design across projects.

ACHIEVEMENTS & CERTIFICATIONS

- 🏆 Hackathon Winner — PW IOI Noida 2026 (competed against 200+ participants)
- 350+ LeetCode problems solved — strong focus on arrays, trees, DP, graphs, and medium/hard concurrency problems
- Full Stack Web Development — Udemy (Angela Yu), 2025
- Quest for Purpose — IIT Kharagpur, 2024
- Class X: 96% | Class XII: 92.8% — among top performers in branch

EDUCATION

B.Tech — Information Technology 2024 – 2028 (Expected)

Harcourt Butler Technical University (HBTU), Kanpur